

ANTIGRAVITY ARCHITECTURE SPEC

VERCEL READY

NEXT.JS 14+

HYBRID MODEL

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Personal Developer Portfolio — Mohammed Sarook Mohammed Safith ("SAFITH THE VIBE CODER 🤖")

OWNER

Mohammed Safith

DEGREE

B.IT (Hons) CGPA 3.47

TARGET ROLES

Full-Stack / QA / IT Ops

TARGET HOST

safith-portfolio.vercel.app

1 Executive Summary & Objectives

This document details the architectural specification for Mohammed Safith's personal portfolio website. Designed as a high-impact engineering portfolio, it is structured to simultaneously satisfy three core audiences:

AUDIENCE	THEIR PRIMARY GOAL	WHAT THIS PORTFOLIO DELIVERS
Recruiters & Engineering Managers	Assess technical competence for Junior Dev, QA, or IT Operations roles.	Clean architecture, unit testing evidence (Liquid Todo, PeerConnect), clean code, downloadable ATS CV.
Freelance Clients	Verify capability to build end-to-end working web & mobile apps.	Live production apps (PeerConnect on Vercel, Daily Books Play Store bundle, MedExpense).
Scholarship & Academic Boards	Review academic excellence, rigor, and research potential.	City University Malaysia degree credentials, CGPA 3.47, verified recommendation letters.

💡 Core Architecture Philosophy

The portfolio itself is built as an **active demonstration of UI engineering skills**. Rather than a static resume page, features like the 4-theme system, multi-language engine with RTL, and interactive project case studies actively prove Safith's hands-on abilities to interviewers.

2 Personal Branding & Identity

Full Legal Name	Mohammed Sarook Mohammed Safith
Display Name & Subtitle	SAFITH — <i>THE VIBE CODER</i> 🤖
Dynamic Title (Hero)	Typed text effect cycling through: Junior Full-Stack Developer · QA & Test Automation Enthusiast · IT Operations Specialist
Location & Education	Colombo / Trincomalee, Sri Lanka 🇱🇰 B.IT (Honours), City University Malaysia
GitHub & Live References	GitHub: github.com/safith001 Flagship Live App: peer-connect-node.vercel.app

3 Technology Stack & Rationale

LAYER	TECHNOLOGY	WHY THIS TOOL WAS CHOSEN (DEFENSIBILITY)
Framework	Next.js 14+ (App Router)	Reinforces existing Next.js skills from PeerConnect. Provides fast SSG/SSR, native SEO optimization, and automatic Vercel deployment.
Styling	Tailwind CSS 3	Utility-first styling mapped to CSS Custom Properties (var(--bg-...)). Enables instantaneous theme switching without DOM rebuilding.
Motion	Framer Motion	Declarative scroll reveals, interactive card tilts, and smooth theme cross-fades with hardware acceleration.
i18n	next-intl	First-class App Router localization with full Right-to-Left (RTL) layout switching for Urdu and Arabic.
Contact Channel	Formspree / EmailJS	Serverless form submission delivering inquiries directly to email without requiring a backend or database.
Deployment	Vercel (Free Tier)	Zero-config continuous deployment from GitHub with edge caching and automatic preview branches.

4 The 4 – Theme Design System Engine

The site features a sticky theme selector that dynamically applies CSS variables on the `<html data-theme="...">` root element. This demonstrates advanced CSS architecture without breaking HTML structure.

1. Glassmorphism Dark

Default Mode: Slate-950 backdrop, frosted glass cards (blur 16px), indigo/violet gradient glows.

Matches PeerConnect aesthetic.

2. Minimal Light

Recruiter Mode: Crisp pure white (#FFF), subtle elevation shadows, blue-600 accents, high contrast.

ATS & corporate-friendly.

3. Warm Earthy

Editorial Mode: Warm cream (#FAF8F5) background, terracotta (#C2684E) & sage green accents.

Approachable craftsmanship.

4. Terminal Hacker

Developer Mode: Deep black (#050505), phosphor matrix green (#00FF41), JetBrains Mono typography.

Shows coding passion.

5 Site Blueprint (Hybrid Architecture)

The site combines a smooth-scrolling single page for rapid recruiter evaluation with dedicated deep-dive case study routes (/projects/[slug]) for technical defensibility:

SECTION	KEY ELEMENTS	USER FLOW & BEHAVIOR
1. Hero	Full name, "Vibe Coder" badge, typed text role cyclcr, 3 primary CTA buttons.	Immediate visual hook; offers one-click scroll to Projects, CV download, or Contact.
2. About Me	City University Malaysia background, CGPA 3.47 stat, continuous learning journey.	Provides academic proof and establishes passion for both QA and full-stack development.
3. Projects	4 interactive cards (PeerConnect, Daily Books, MedExpense, Liquid Todo).	Displays tech chips, live demo links, GitHub links, and link to full case study.
4. Skills Matrix	Languages, Frontend, Backend, Mobile, Databases, QA/DevOps groupings.	Structured visual pill badges matching target job descriptions.
5. Experience	Degree timeline, Guildford Integrated Systems internship, project milestones.	Validates 120 credit hours and director recommendation letter.
6. Contact & CV	Formspree form (Name, Email, Message) + direct PDF resume download button.	Zero friction for recruiters or clients to initiate inquiries.
7. Case Studies	/projects/[slug] dedicated pages with architecture diagrams and lessons.	Allows technical interviewers to inspect the exact engineering choices.

6 Flagship Project Breakdown

1. PeerConnect — Campus Social Networking Platform LIVE ON VERCEL A production social network connecting university students with email verification, real-time Firestore chat, academic feed, and Cloudinary PDF slide sharing. Next.js 16 Firebase Auth Cloud Firestore Cloudinary Tailwind CSS TypeScript
2. Daily Books — Tamil Islamic PDF Library & Habit Tracker PLAY STORE READY An elder-friendly, 100% offline PDF reader app with 100 curated books, multi-touch zoom engine, 7-day streak tracker, and decoupled Node.js REST API backend. Node.js API Express Capacitor Flutter Android Gradle
3. MedExpense — MBBS Smart Spend Tracker with Gemini AI MOBILE APP Offline-first Flutter expense ledger with SQLite, automated month-end background sync to Google Sheets, and financial coaching powered by Google Gemini 2.5 Flash. Flutter / Dart SQLite Google Apps Script Gemini AI API
4. Liquid Todo — Task Management & QA Testing Suite QA & TDD FOCUS Relational task manager with persistent SQLite storage and an automated test suite verifying database transactions and schema constraints. Python Flask SQLite Integration Testing

7 Multilingual & Internationalization Engine

The site is architected for global accessibility with route-based localization (`/[locale]/...`) managed by `next-intl`.

LANGUAGE	CODE	DIRECTION	LAUNCH PHASE	SPECIAL ARCHITECTURAL REQUIREMENTS
English	en	LTR	Tier 1 (Launch)	Default base translation file (<code>messages/en.json</code>).
Tamil	ta	LTR	Tier 1 (Launch)	Unicode Noto Sans Tamil typography support.
Urdu	ur	RTL	Tier 1 (Launch)	Right-to-Left layout switching, CSS logical properties (<code>ms-*</code> , <code>me-*</code>).
Sinhala, Hindi, German, Russian, etc.	9 codes	LTR/RTL	Tier 2 (Roadmap)	Decoupled JSON dictionaries allow seamless addition without code changes.

8 Phased Sprint Implementation Roadmap

Sprint 1: Foundation, Design Tokens & Core Layout ▶ Initialize Next.js 14+ project with TypeScript and Tailwind CSS. ▶ Configure CSS custom properties for all 4 themes in <code>globals.css</code>. ▶ Build responsive Header with sticky theme dropdown and mobile navigation. ▶ Build <code>HeroSection</code> with typed text animation and primary CTA buttons. ▶ Establish continuous deployment pipeline on Vercel.	WEEK 1
Sprint 2: Content Modules & Motion Choreography ▶ Implement <code>AboutSection</code> with academic metrics (CGPA 3.47, 4+ apps). ▶ Implement <code>ProjectsSection</code> with responsive grid and <code>ProjectCard</code> components. ▶ Implement <code>SkillsSection</code> with categorized domain groups. ▶ Implement <code>TimelineSection</code> documenting degree and internship milestones. ▶ Add Framer Motion scroll reveals (<code>whileInView</code>) across all sections.	WEEK 2
Sprint 3: Case Studies, Contact Form & Resume Access ▶ Create dynamic route <code>/projects/[slug]</code> with shared case study layout. ▶ Write technical case study content for PeerConnect, Daily Books, MedExpense, and Todo. ▶ Implement <code>ContactSection</code> with Formspree headless email endpoint. ▶ Wire verified PDF resume download button linked to <code>public/cv/</code>. ▶ Integrate active section highlighting via <code>useScrollSpy</code>.	WEEK 3
Sprint 4: Multilingual Engine, RTL & Production Launch ▶ Set up <code>next-intl</code> route localization (<code>/[locale]/...</code>). ▶ Populate English, Tamil, and Urdu dictionaries in <code>messages/</code>. ▶ Enforce RTL styling rules and mirror layouts for Urdu locale. ▶ Perform Lighthouse audit (Target ≥ 90 on Performance, A11y, Best Practices, SEO). ▶ Production release on Vercel with Open Graph social metadata.	WEEK 4

9 Quality Assurance & Acceptance Criteria

QUALITY VECTOR	ACCEPTANCE STANDARD
Build Integrity	Zero errors during <code>npx next build</code> ; strict TypeScript type checking enabled.
Performance Score	Lighthouse score ≥ 90 across Performance, Accessibility, Best Practices, and SEO.
Responsiveness	Pixel-perfect rendering across Mobile (375px+), Tablet (768px+), and Desktop (1280px+).
Theme Reliability	All 4 themes transition cleanly under 300ms with zero contrast violations.
RTL Symmetry	Urdu layout mirrors correctly with proper text alignments and icon directions.
Contact Inquiries	Submitting contact form successfully dispatches an email notification to Safith.